# Class 13: Transcriptomics and the analysis of RNA-Seq data

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### Background

Today we will perform an RNASeq analysis on the effects of dexamethasone (hereafter "dex") a common steroid, on airway smooth muscle (ASM) cell lines.

### Import/ Read the data from Himes et al.

```
counts <- read.csv("~/BIMM 143/class13/airway_scaledcounts.csv",
                   row.names = 1)
metadata <- read.csv("~/BIMM 143/class13/airway_metadata.csv")
```

Let's have a peak at this data

```
head(metadata)
```

```
          id     dex celltype     geo_id
1 SRR1039508 control   N61311 GSM1275862
2 SRR1039509 treated   N61311 GSM1275863
3 SRR1039512 control  N052611 GSM1275866
4 SRR1039513 treated  N052611 GSM1275867
5 SRR1039516 control  N080611 GSM1275870
6 SRR1039517 treated  N080611 GSM1275871
```

Sanity check corespondence of counts and metadata

```
metadata$id
```

```
[1] "SRR1039508" "SRR1039509" "SRR1039512" "SRR1039513" "SRR1039516"
[6] "SRR1039517" "SRR1039520" "SRR1039521"
```

```
colnames(counts)
```

```
[1] "SRR1039508" "SRR1039509" "SRR1039512" "SRR1039513" "SRR1039516"
[6] "SRR1039517" "SRR1039520" "SRR1039521"
```

```
all( metadata$id == colnames(counts))
```

```
[1] TRUE
```

Q1. How many genes are in this dataset?

There are 38694 genes in this dataset.

Q2. How many 'control' cell lines do we have?

```
n.control <- sum(metadata$sex == "control")
```

There are 0 control cell lines in this dataset.

### Extract and summarize the control samples

To find out where the control samples are we need the metadata

Q3. How would you make the above code more robust?

```
control <- metadata[metadata$dex == "control", ]
control.counts <- counts[ , control$id]
control.mean <- rowMeans(control.counts)
head(control.mean)
```

```
ENSG00000000003 ENSG00000000005 ENSG00000000419 ENSG00000000457 ENSG00000000460
         900.75            0.00          520.50          339.75           97.25
ENSG00000000938
           0.75
```

### Extract and summarize the treated (i.e. drug) samples

Q4. Calculate the mean counts for treated samples

```
treated <- metadata[metadata$dex == "treated", ]
treated.counts <- counts[ , treated$id]
treated.mean <- rowMeans(treated.counts)
```

Store these results together in a new data frame called `meancounts`

```
meancounts <- data.frame (control.mean, treated.mean)
```

Q5(a). What plot is used?

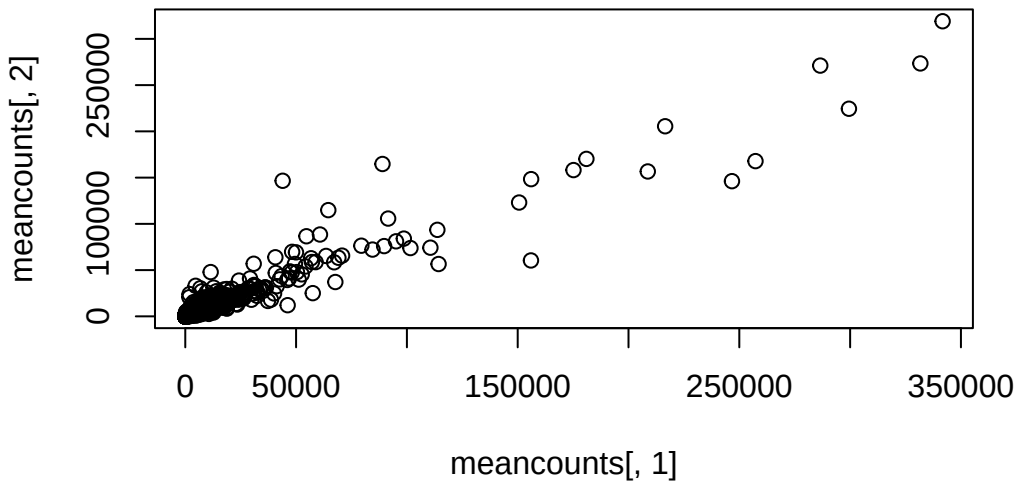
Q5(b). What ggplot geom is used?

Geom_point()

Scatter plot of treated mean vs control mean. A basic plot() scatter plot.

Now let's make a plot to explore these results

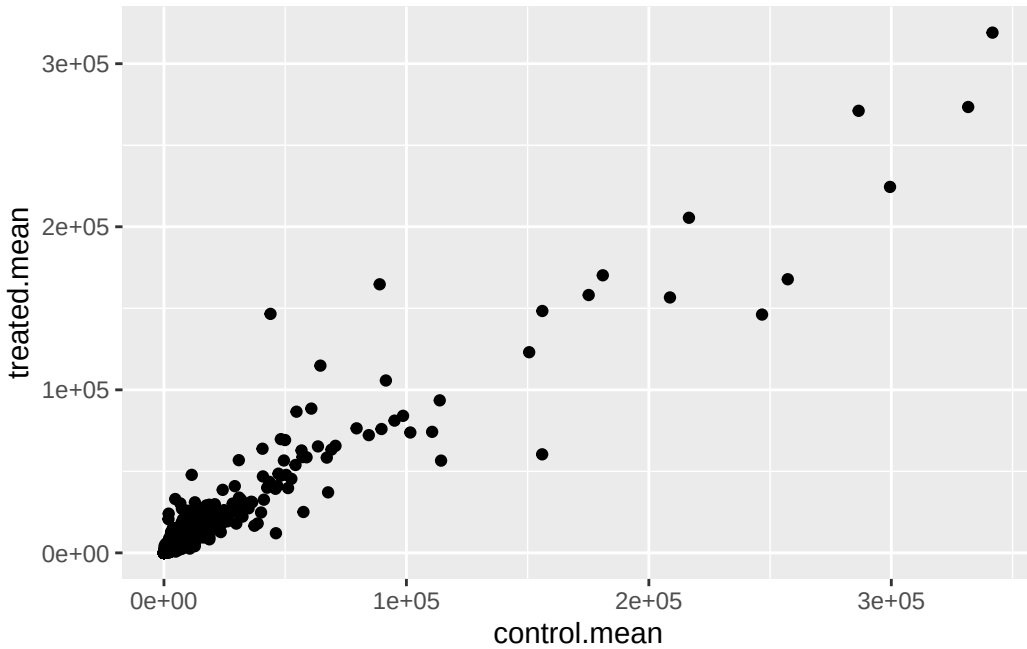
```
plot(meancounts[ ,1], meancounts[ ,2])
```




```
library(ggplot2)
```

```
Warning: package 'ggplot2' was built under R version 4.5.2
```

```
ggplot(meancounts) +
  aes(control.mean, treated.mean) +
  geom_point()
```




Q6. What argument in plot() puts axes on a log scale?

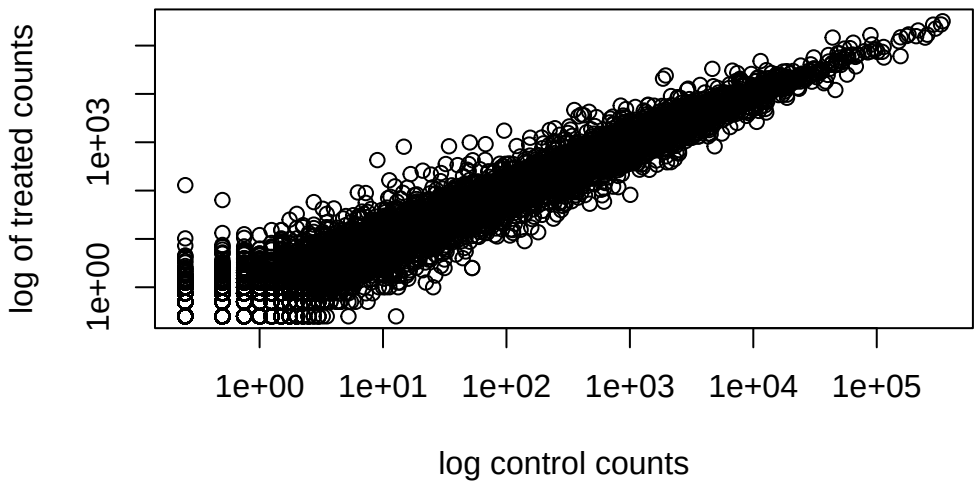
log = "xy"

We will make a log-log plot to draw out this skewed data and see what is going on.

```
plot(meancounts[ ,1], meancounts[ ,2], log="xy",
     xlab= "log control counts",
     ylab= "log of treated counts")
```

```
Warning in xy.coords(x, y, xlabel, ylabel, log): 15032 x values <= 0 omitted
from logarithmic plot

Warning in xy.coords(x, y, xlabel, ylabel, log): 15281 y values <= 0 omitted
from logarithmic plot
```




We often log2 transformations when dealing with this sort of data.

```
log2(20/20)
```

```
[1] 0
```

```
log2(40/20)
```

```
[1] 1
```

```
log2(20/40)
```

```
[1] -1
```

```
log2(80/20)
```

```
[1] 2
```

This log2 transformation has this property where if there is no change the log2 value will be zero and if it is doubled the value will be 1 and if it is halved it will be -1 and so on.

So let's add a log2 fold change column to our results so far.

```
meancounts$log2fc <- log2(meancounts$treated.mean /
                          meancounts$control.mean)
```

```
head(meancounts)
```

```
                control.mean treated.mean      log2fc
ENSG00000000003       900.75       658.00 -0.45303916
ENSG00000000005         0.00         0.00         NaN
ENSG00000000419       520.50       546.00  0.06900279
ENSG00000000457       339.75       316.50 -0.10226805
ENSG00000000460        97.25        78.75 -0.30441833
ENSG00000000938         0.75         0.00        -Inf
```

We need to get rid of zero count genes that we can not say anything about

> Q7. Purpose of arr.ind = TRUE + using first column + unique()

```
zero.vales <- which( meancounts[, 1:2]==0, arr.ind = TRUE)
to.rm <- unique(zero.vales[,1])
mycounts <- meancounts[-to.rm, ]
```

```
head(mycounts)
```

```
                control.mean treated.mean      log2fc
ENSG00000000003       900.75       658.00 -0.45303916
ENSG00000000419       520.50       546.00  0.06900279
ENSG00000000457       339.75       316.50 -0.10226805
ENSG00000000460        97.25        78.75 -0.30441833
ENSG00000000971      5219.00      6687.50  0.35769358
ENSG00000001036      2327.00      1785.75 -0.38194109
```

How many genes are remaining?

```
nrow(mycounts)
```

```
[1] 21817
```

### Use fold change to see up and down regulated genes.

A common threshold used for calling something differentially expressed is a log2(FoldChange) of greater than 2 or less than -2. Let's filter the dataset both ways to see how many genes are up or down-regulated.

> Q8

```
sum(mycounts$log2fc > 2)
```

```
[1] 250
```

> Q9

and down regulated

```
sum(mycounts$log2fc < -2)
```

```
[1] 367
```

> Q10 Do we trust these results? Not fully because we don't know if these changes are significant yet

### DESeq2 analysis

Let's do this the right way. DESeq2 is an R package specifically for analyzing count-based NGS data like RNA-seq. It is available from Bioconductor.

```
 # load up DESeq2
library(DESeq2)
```

```
Warning: package 'DESeq2' was built under R version 4.5.2

Loading required package: S4Vectors

Warning: package 'S4Vectors' was built under R version 4.5.2

Loading required package: stats4

Loading required package: BiocGenerics
```

```
Warning: package 'BiocGenerics' was built under R version 4.5.2

Loading required package: generics


Attaching package: 'generics'

The following objects are masked from 'package:base':

    as.difftime, as.factor, as.ordered, intersect, is.element, setdiff,
    setequal, union


Attaching package: 'BiocGenerics'

The following objects are masked from 'package:stats':

    IQR, mad, sd, var, xtabs

The following objects are masked from 'package:base':

    anyDuplicated, aperm, append, as.data.frame, basename, cbind,
    colnames, dirname, do.call, duplicated, eval, evalq, Filter, Find,
    get, grep, grepl, is.unsorted, lapply, Map, mapply, match, mget,
    order, paste, pmax, pmax.int, pmin, pmin.int, Position, rank,
    rbind, Reduce, rownames, sapply, saveRDS, table, tapply, unique,
    unsplit, which.max, which.min


Attaching package: 'S4Vectors'

The following object is masked from 'package:utils':

    findMatches

The following objects are masked from 'package:base':

    expand.grid, I, unname

Loading required package: IRanges
```

```
Warning: package 'IRanges' was built under R version 4.5.2

Loading required package: GenomicRanges

Warning: package 'GenomicRanges' was built under R version 4.5.2

Loading required package: Seqinfo

Warning: package 'Seqinfo' was built under R version 4.5.2

Loading required package: SummarizedExperiment

Warning: package 'SummarizedExperiment' was built under R version 4.5.2

Loading required package: MatrixGenerics

Warning: package 'MatrixGenerics' was built under R version 4.5.2

Loading required package: matrixStats


Attaching package: 'MatrixGenerics'

The following objects are masked from 'package:matrixStats':

    colAlls, colAnyNAs, colAnys, colAvgsPerRowSet, colCollapse,
    colCounts, colCummaxs, colCummins, colCumprods, colCumsums,
    colDiffs, colIQRDiffs, colIQRs, colLogSumExps, colMadDiffs,
    colMads, colMaxs, colMeans2, colMedians, colMins, colOrderStats,
    colProds, colQuantiles, colRanges, colRanks, colSdDiffs, colSds,
    colSums2, colTabulates, colVarDiffs, colVars, colWeightedMads,
    colWeightedMeans, colWeightedMedians, colWeightedSds,
    colWeightedVars, rowAlls, rowAnyNAs, rowAnys, rowAvgsPerColSet,
    rowCollapse, rowCounts, rowCummaxs, rowCummins, rowCumprods,
    rowCumsums, rowDiffs, rowIQRDiffs, rowIQRs, rowLogSumExps,
    rowMadDiffs, rowMads, rowMaxs, rowMeans2, rowMedians, rowMins,
    rowOrderStats, rowProds, rowQuantiles, rowRanges, rowRanks,
    rowSdDiffs, rowSds, rowSums2, rowTabulates, rowVarDiffs, rowVars,
    rowWeightedMads, rowWeightedMeans, rowWeightedMedians,
    rowWeightedSds, rowWeightedVars
```

```
Loading required package: Biobase

Warning: package 'Biobase' was built under R version 4.5.2

Welcome to Bioconductor

    Vignettes contain introductory material; view with
    'browseVignettes()'. To cite Bioconductor, see
    'citation("Biobase")', and for packages 'citation("pkgname")'.


Attaching package: 'Biobase'

The following object is masked from 'package:MatrixGenerics':

    rowMedians

The following objects are masked from 'package:matrixStats':

    anyMissing, rowMedians
```

```r
dds <- DESeqDataSetFromMatrix(countData=counts,
                              colData=metadata,
                              design=~dex)
```

```
converting counts to integer mode

Warning in DESeqDataSet(se, design = design, ignoreRank): some variables in
design formula are characters, converting to factors
```

```r
dds <- DESeq(dds)
```

```
estimating size factors

estimating dispersions

gene-wise dispersion estimates
```

```
mean-dispersion relationship

final dispersion estimates

fitting model and testing
```

```r
res <- results(dds)
res
```

```
log2 fold change (MLE): dex treated vs control
Wald test p-value: dex treated vs control
DataFrame with 38694 rows and 6 columns
                 baseMean log2FoldChange     lfcSE      stat    pvalue
                <numeric>      <numeric> <numeric> <numeric> <numeric>
ENSG00000000003  747.1942      -0.350703  0.168242 -2.084514 0.0371134
ENSG00000000005    0.0000             NA        NA        NA        NA
ENSG00000000419  520.1342       0.206107  0.101042  2.039828 0.0413675
ENSG00000000457  322.6648       0.024527  0.145134  0.168996 0.8658000
ENSG00000000460   87.6826      -0.147143  0.256995 -0.572550 0.5669497
...                   ...            ...       ...       ...       ...
ENSG00000283115  0.000000             NA        NA        NA        NA
ENSG00000283116  0.000000             NA        NA        NA        NA
ENSG00000283119  0.000000             NA        NA        NA        NA
ENSG00000283120  0.974916       -0.66825   1.69441 -0.394385  0.693297
ENSG00000283123  0.000000             NA        NA        NA        NA
                     padj
                <numeric>
ENSG00000000003  0.163017
ENSG00000000005        NA
ENSG00000000419  0.175937
ENSG00000000457  0.961682
ENSG00000000460  0.815805
...                   ...
ENSG00000283115        NA
ENSG00000283116        NA
ENSG00000283119        NA
ENSG00000283120        NA
ENSG00000283123        NA
```

We can get some basic summary tallies using the `summary()` function

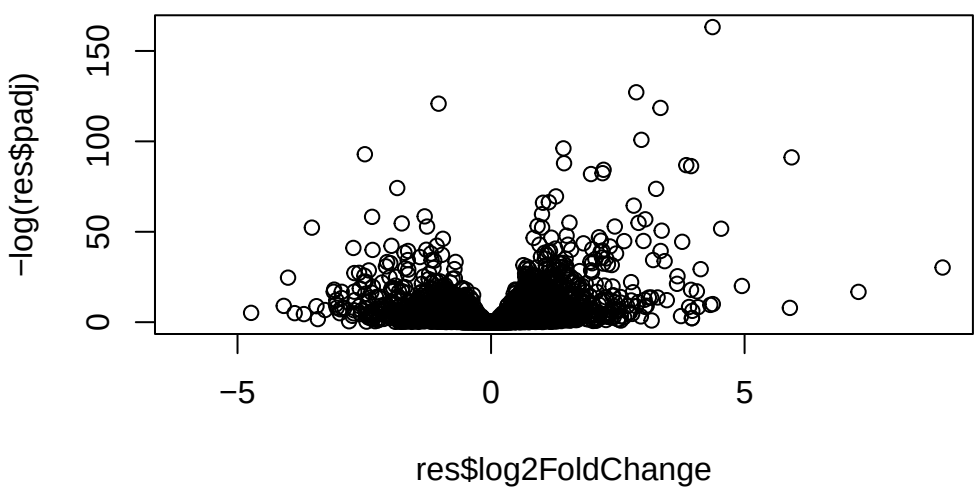
```r
summary(res, alpha=0.05)
```

```
out of 25258 with nonzero total read count
adjusted p-value < 0.05
LFC > 0 (up)       : 1242, 4.9%
LFC < 0 (down)     : 939, 3.7%
outliers [1]       : 142, 0.56%
low counts [2]     : 9971, 39%
(mean count < 10)
[1] see 'cooksCutoff' argument of ?results
[2] see 'independentFiltering' argument of ?results
```

### Volcano plot

Make a summary plot of our results.

```r
plot(res$log2FoldChange, -log(res$padj))
```



```r
log(0.1)
```

```
[1] -2.302585
```

```r
log(0.005)
```

```
[1] -5.298317
```

Finish for today by saving our results

```r
write.csv(res, file="DESeq2_results.csv")
```